Article

fp-tools: A Reproducible Platform for ATAC-seq Footprinting and Regulatory Motif Analysis

Yaoxiang Li [1] and Chunling Yi [1,*]

[1] Department of Oncology, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Georgetown University, Washington, DC, USA; yl814@georgetown.edu (Y.L.); cy232@georgetown.edu (C.Y.)
* Correspondence: cy232@georgetown.edu (C.Y.)

Abstract

ATAC-seq footprinting enables near-base-resolution inference of transcription-factor (TF) occupancy across the genome. However, the field lacks a computationally efficient, user-friendly, and integrated platform that accepts both bulk and single-cell ATAC-seq data, supports both de novo motif discovery and matching to public motif databases, and generates interactive and publication-ready visual outputs. Here, we present `fp-tools`, a Python package that builds on the widely used ATAC-seq footprinting framework TOBIAS. We substantially improve computational efficiency across key steps, including Tn5 bias correction, footprint calling, and motif matching. In addition, `fp-tools` introduces de novo motif discovery, replicate-aware differential footprinting, and aggregation analysis, features not provided as native TOBIAS workflows. We further streamline a single-cell ATAC-seq footprinting workflow, provide interactive visualization of analysis results, and implement a graphical user interface (GUI) that enables code-free execution of all package commands. Using public bulk ATAC-seq replicates and single-cell datasets, we demonstrate that `fp-tools` quantitatively recovers cell-type-specific TF-binding patterns and identifies de novo binding motifs that are not captured by existing public motif databases. Together, `fp-tools` provides an efficient, accessible, and comprehensive platform for TF footprint analysis across bulk and single-cell ATAC-seq studies.

Keywords: ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; pseudobulk ATAC-seq; replicate-aware reporting; reproducibility; command-line bioinformatics

1. Introduction

Chromatin accessibility profiling by the assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become a standard readout of active regulatory DNA [1]. Within accessible regions, sequence-specific transcription factors (TFs) can protect bound DNA from transposition, producing local cut-site depletions, or "footprints," with increased accessibility in nearby flanks. Detecting these motif-centered cut-site patterns supports inference of TF occupancy, differential regulatory activity between conditions, and interpretation of non-coding regulatory variants. Earlier footprinting and occupancy models, including CENTIPEDE, Wellington, PIQ, and DeFCoM, established the value of combining sequence information with chromatin accessibility profile shape for TF-binding-site inference [2–5].

Despite this progress, practical ATAC-seq footprinting remains difficult to run as a complete and reproducible analysis. Widely used tools such as TOBIAS [6] and HINT-ATAC [7] provide enzymatic bias correction and footprint scoring for bulk ATAC-seq data.

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Multiscale and nucleosome-aware approaches, including the PRINT/scPrinter family [8], extend footprint modeling across different scales of chromatin protection. Supervised TF-binding-site predictors such as maxATAC [9] and sequence-to-profile models such as ChromBPNet [10] provide complementary prediction frameworks. Motif-discovery and motif-comparison tools, including MEME/STREME [11,12] and Tomtom [13], are commonly used after peak or footprint selection, and curated databases such as JASPAR [14] remain central for known-motif matching. These tools are valuable, but they are typically assembled through separate command interfaces, file conventions, report formats, and visualization steps.

This fragmentation creates a practical barrier for many applied studies. A complete analysis must connect ATAC-seq input processing, Tn5 sequence-bias correction, footprint scoring, motif matching, optional de novo motif discovery, differential footprint analysis, aggregate visualization, and publication-ready reporting. Single-cell ATAC-seq adds another difficulty: sparse per-cell fragments usually need to be aggregated into cell-type, cluster, or neighborhood-level profiles before footprinting can be interpreted. For many biology laboratories, the limiting step is therefore not one missing algorithm, but the absence of an efficient platform that connects these steps and preserves the analysis record.

We developed `fp-tools` to address this need. The package builds on the widely used TOBIAS ATAC-seq footprinting framework and keeps its interpretable motif-centered footprint logic, while improving speed and usability around the core analysis. Compared with TOBIAS, `fp-tools` adds de novo motif preparation and matching, replicate-aware differential footprint reports, pseudobulk single-cell ATAC-seq support, interactive HTML reports, editable SVG outputs, and a graphical user interface for code-free execution of the same command-compatible workflows.

Here, we describe the `fp-tools` implementation and evaluate it with public bulk ATAC-seq replicates and public single-cell ATAC-seq datasets. We focus on practical improvements to established ATAC-seq footprinting: connecting common TF-footprinting tasks into one auditable analysis that can be run from the command line, saved YAML files, or the GUI.

2. Materials and Methods

2.1. Package Architecture and Core Workflows

`fp-tools` accelerates the main signal-processing steps used in ATAC-seq footprinting. The core analysis includes Tn5 sequence-bias correction, footprint scoring over corrected signal, motif-anchored differential analysis, and aggregate visualization around TFBS or region sets. These steps use shared region, signal, motif, and sample-label conventions, so the output of one step can be used directly by the next. YAML configuration files provide a reproducible way to save and rerun the same analysis without changing the underlying methods.

Following the signal model popularized by TOBIAS [6], `fp-tools` treats base-resolution, bias-corrected cut-site profiles as the primary footprinting signal. This design preserves the familiar bulk ATAC footprinting interpretation: the biological signal is a corrected center-versus-flank cut-site contrast at motif or candidate sites. After bias correction, `fp-tools` can additionally rescale corrected cut-site tracks before footprint scoring by matching the 95th percentile of signal over shared background regions to an across-sample target, usually the median. This robust q95 scaling reduces sample-to-sample signal-level variation before footprint scoring while being less sensitive to rare extreme bins than full quantile matching. The package keeps this signal definition while recording sample labels, input manifests, reusable configuration, replicate summaries, and report outputs. For footprint calling, let x_i denote corrected Tn5 cut-site signal around candidate position p. The default footprint score is a center-versus-flank contrast,

$$F(p) = \frac{1}{|L| + |R|}\left(\sum_{i \in L} x_i + \sum_{i \in R} x_i\right) - \frac{1}{|C|}\sum_{i \in C} x_i,$$

where C is the protected center and L and R are flanking windows. Larger values therefore correspond to the expected footprint pattern: fewer cuts at the putative bound site and more cuts in nearby accessible flanks.

2.2. De Novo Motif Discovery Preparation

The de novo motif-discovery module begins from candidate footprint intervals, which may be produced by footprint calling or provided as input. For each candidate interval j with center p_j, strand a_j, and flank size h, it exports a candidate-centered sequence

$$S_j = G[p_j - h, p_j + h],$$

reverse-complementing the sequence when strand information indicates the negative strand. The module records the input intervals, genome build, flank width, software versions, and MEME, DREME, STREME [11,12,15] or optional Tomtom [13] settings needed to reproduce each run. STREME was designed for accurate motif discovery from enriched sequence sets [12]. Discovered motifs can be analyzed as an independent motif set or matched to a known database such as JASPAR 2026 [14] and appended to the known motif set for downstream differential footprint analysis. In this design, candidate FASTA export, motif discovery, motif comparison, and database augmentation become part of the recorded footprint analysis.

2.3. Pseudobulk Fragment Aggregation

The pseudobulk module groups single-cell ATAC fragments by a metadata column, such as cell type, cluster, donor, or treatment. Let F_b be the multiset of fragments assigned to barcode b, and let $g(b)$ be its group label. The pseudobulk fragment collection for group k is the concatenated multiset

$$F_k = \mathrm{concat}_{b:g(b)=k}(F_b),$$

so repeated fragments and fragment multiplicity are preserved. Groups that pass specified cell-count and fragment-count filters are written as bulk-like fragment files with manifest records. When genome sizes are supplied, `fp-tools` can also emit CPM-normalized cut-site bigWigs,

$$y_{k,t} = 10^6 \frac{c_{k,t}}{\sum_u c_{k,u}},$$

where $c_{k,t}$ is the cut-site count for group k at genomic position t. These tracks support motif-centered aggregate profiles matched to the bulk ATAC-seq analysis path. The full pseudobulk route writes pseudo-paired BAMs from grouped fragments and then runs correction, scoring, motif-aware comparison, and plotting with the same conventions used for bulk ATAC-seq.

For cell-level marker visualization, `fp-tools` also computes K-nearest-neighbor (KNN)-smoothed footprint signatures from single-cell fragments. For each marker TF, motif centers are scanned inside the selected accessible-bin universe. Fragment cut sites are counted in a $\pm$100 bp window around those motif centers for each barcode. Each cell is assigned a local neighborhood profile by summing its 75 nearest neighbors in the single-cell embedding, and a flank-minus-center footprint-signature score is computed from that neighborhood profile. Scores are oriented so that the expected marker cell type has the positive direction and then standardized per TF for UMAP display. These KNN-smoothed signatures summarize local, motif-centered cut-site patterns across neighboring cells, while full ATAC-corrected footprint tracks are computed from the grouped pseudobulk profiles.

2.4. Replicate-aware Differential Footprint Analysis

The differential footprint module supports both single-sample contrasts and replicate-supported comparisons. It retains conventional motif-centered footprint-score logic while adding repeated condition labels, replicate-level summaries, and replicate-aware testing. Repeated condition names, for example `-cond-names Bcell Bcell Tcell Tcell`, define biological replicate groups while preserving the original differential-footprint output structure. Let $x_{c,r,i}$ be the footprint score at site or background position i for replicate r of condition c.

For one-sample versus one-sample contrasts, `diff-footprints` uses the motif-site versus background-distribution comparison of the standard TOBIAS-style report. Candidate sites are grouped by motif, site-level changes at motif sites are compared with matched background changes, and p-values are adjusted across motifs within each contrast.

For replicate-supported comparisons, `fp-tools` generalizes this comparison to multiple samples per condition. Footprint scores are computed for each replicate on the same peak and motif-site universe; motif-site and background distributions are then summarized at the condition level before reporting the same change, log-fold-change, p-value, and adjusted q-value columns. The report also includes replicate counts, condition means, SDs, and SE summaries, so the additional samples contribute to a more stable estimate rather than being collapsed outside the analysis.

Quantile normalization remains an optional comparison-stage adjustment. In single-sample contrasts, sample-quantile normalization provides the conventional two-sample score-distribution matching. In replicate-supported comparisons, condition-quantile normalization averages background distributions within each condition, fits one normalization factor per condition, and applies that factor to the replicates before condition summaries are computed. This option is used as a sensitivity adjustment; the main replicate-aware extension is the explicit use of replicate-level motif scores.

When per-sample motif-score matrices are available, `fp-tools` further applies empirical-Bayes variance moderation across motifs, following the moderated testing framework of Smyth [16]. This replicate-supported view estimates condition effects from the per-sample motif scores and moderates the per-motif variance across the motif set before computing adjusted differential-footprint statistics.

2.5. Public Data Processing and Evaluation

The bulk replicate demonstration uses a 3-vs-3 public ENCODE ATAC-seq comparison between K562 and HepG2. We used released GRCh38 alignment BAMs from K562 ENCSR868FGK (ENCFF077FBI, ENCFF128WZG, and ENCFF534DCE) and HepG2 ENCSR291GJU (ENCFF624SON, ENCFF926KFU, and ENCFF990VCP). Problematic genomic regions were filtered using the ENCODE blacklist [17]. Replicate peak intervals were merged into a common peak universe, bias-corrected, q95-scaled over shared background regions, footprint-called, and analyzed by replicate-aware differential footprint comparison. The analysis used JASPAR 2026 CORE vertebrates non-redundant motifs [14] for known-motif scanning. Exact command-line parameters and software versions are recorded in source TSV files in the manuscript `tables` directory and summarized in Tables S1 and S2.

Public inputs are organized as versioned TSV manifests that record source URLs, accessions, assemblies, checksums, local paths, and analysis labels. Discovery and download scripts query public repositories such as ENCODE [18] for released human GRCh38 ATAC-seq records and write resumable download reports. The manuscript evaluation combines compact regression inputs, public bulk ATAC replicates, de novo motif-preparation examples, and public PBMC single-cell ATAC examples. For each analysis, command-line parameters, software versions, resolved input manifests, source tables, and generated figure inputs are kept with the manuscript resources so the reported examples can be rerun and audited.

2.6. Reproducibility Engineering

The package includes unit tests for numerical helpers and CLI behavior, plus automated output checks for footprint calling, differential footprint analysis, and aggregate plotting, with an opt-in slow check for bias correction. Continuous integration runs the suite on compact reference inputs, builds source and wheel artifacts, and verifies the installed console scripts. The repository also includes `environment.yml`, a Dockerfile, release metadata, and Make targets for testing, building, and manuscript checks. These choices follow FAIR-oriented research-software principles and established bioinformatics environment practices [19–21]. Analysis scripts record random seeds, tool versions, command lines, and resolved manifests alongside the corresponding outputs. Figure source tables and editable figure files are retained with the manuscript resources for auditability.

3. Results

We evaluated `fp-tools` for command stability, local runtime and memory behavior, and biological plausibility in public bulk and single-cell ATAC-seq examples.

3.1. A Unified, Reproducible Footprinting Platform

`fp-tools` consolidates the core ATAC-seq footprinting steps into one command-compatible analysis. Bias correction, footprint calling, differential footprint analysis, and aggregate visualization share region and signal conventions, reducing the need for file-format conversion between steps. YAML configuration records the same analysis in a machine-readable form for reruns and batch processing (Figure 1).

3.2. Interactive Interfaces and Report Review

In addition to direct command-line execution, `fp-tools` provides browser-based interfaces for configuring analyses and reviewing results. The GUI writes reusable YAML configurations that remain compatible with command-line execution and provides guided panels for the same core commands shown in the documentation (Figure 6). Differential-footprint and aggregate-batch runs also produce self-contained HTML reports with embedded compressed data, searchable motif controls, editable group colors, synchronized aggregate profiles, and SVG export for publication-ready editing.

3.3. Tested Output Stability

The package includes automated checks that run core commands on compact reference inputs and compare key summary outputs for footprint calling, differential footprint analysis, and aggregate plotting. Additional checks exercise public command-line entry points and saved YAML configurations. Together, these tests help ensure that updates preserve the expected behavior of the main workflows and shared input/output conventions.

3.4. De Novo Motif Discovery Complements Database-based Motif Analysis

The de novo motif-preparation path converts candidate footprint intervals into centered FASTA files, runs MEME, DREME, or STREME, optionally compares discovered motifs with a known database using Tomtom, and writes TSV/HTML summaries with

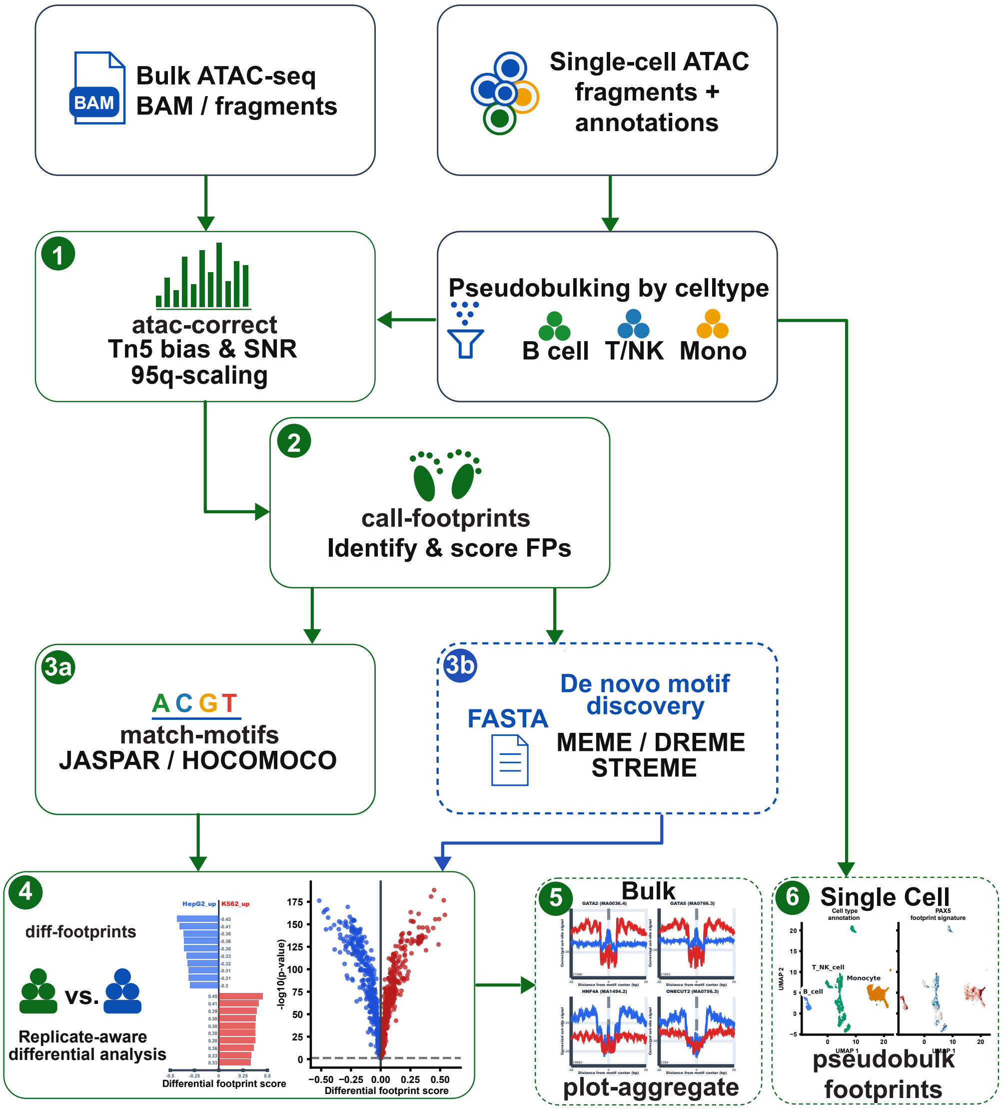

Figure 1. Overview of the `fp-tools` ATAC-seq footprinting platform. Bulk ATAC-seq BAM/fragments and single-cell ATAC fragments with annotations enter a shared command-oriented analysis. The analysis supports Tn5 bias correction, footprint calling, motif matching against curated databases, de novo motif discovery, replicate-aware differential footprint analysis, aggregate footprint visualization, and pseudobulk single-cell footprint analysis.

inline consensus logos. This extends the TOBIAS-style known-motif path by adding a recorded handoff from footprint candidates to motif discovery; it does not change the center-versus-flank footprint scoring model. The module can be run alone to discover recurring patterns in candidate intervals, or after known-motif scanning to find motifs that are absent from the selected database.

We tested whether candidate-derived motifs add information beyond a fixed database using a 3-vs-3 public ENCODE K562 versus HepG2 ATAC-seq comparison [18]. Candidate footprints were called from bias-corrected cut-site tracks, condition-specific candidate FASTA files were exported with 75 bp flanks, and STREME was run reciprocally with K562 candidates against HepG2 controls and HepG2 candidates against K562 controls. These discovered motifs were then used in two modes. In de novo-only mode, the candidate-derived motifs were tested as their own motif set. In supplement mode, the same motifs were appended to a known motif database before running the sample-quantile-normalized differential-footprint analysis.

The full JASPAR 2026 vertebrate non-redundant set contained 1019 motifs, and the reciprocal STREME runs contributed 500 candidate-derived motifs from both ENCODE cell lines. We interpret this run as a site-recovery test of how far de novo-only discovery can cover database-supported footprints and complement a curated motif database. In de novo-only mode, recovery was measured as base-pair overlap after merging the bound-site windows produced by the de novo scan and the corresponding JASPAR-supported scan. By this operational metric, discovered motifs recovered 84.0% of merged JASPAR-supported bound-site-window base pairs across the K562/HepG2 comparison, while 16.0% remained unique to the full JASPAR scan. The de novo-only scan also introduced candidate-derived bound-site windows corresponding to 6.0% of the de novo-only merged window base pairs.

In supplement mode, discovered motifs are appended to known motifs before the same motif-aware differential footprint analysis. As a sensitivity test, we constructed a K562/HepG2-focused restricted JASPAR 2026 set with 867 motifs by removing common erythroid/K562 and liver/HepG2 motif families, then reran the comparison with and without the de novo motifs. In the interactive report, the restricted database yielded 92 display-priority motif families according to the `highlighted` report-selection flag; adding the 500 de novo motifs increased this browsing set to 153 families, including 24 de novo-derived motifs. Because `highlighted` is not a multiple-testing-corrected significance label, these counts are treated only as report-prioritization output. This sensitivity test illustrates the intended use case: when a chosen motif database is incomplete or conservative, candidate-derived motifs can be evaluated alone and can also be carried into the same differential footprint report as database-supported motifs (Figure 2).

A representative K562 versus HepG2 differential footprint report is shown in Figure 3. The report combines selected motif logos, top differential motif rankings, a volcano summary, and replicate-level aggregate footprint profiles in one browser-review view.

3.5. Replicate-aware Differential Footprint Analysis

The replicate-aware analysis reports differential footprint effect sizes, replicate support, and uncertainty summaries while keeping each comparison anchored to a common peak and motif-site universe. In the K562 versus HepG2 3-vs-3 analysis, all six replicates were scanned on the merged ENCODE peak set before comparison, so no additional peak alignment step was needed at the reporting stage.

Repeated condition names define biological replicate groups. In one-sample contrasts, the report uses the motif-site versus background-distribution comparison. When multiple samples are available, motif scores are first aggregated per sample and then analyzed with empirical-Bayes variance moderation across motifs [16]. The K562/HepG2 replicate-supported output reports condition means, replicate counts, SD and SE summaries, moderated statistics, and adjusted q-values alongside the original motif-background comparison.

3.6. Single-cell ATAC Fragments Support Pseudobulk Footprinting

We evaluated pseudobulk aggregation using the public 10x Genomics 5k PBMC single-cell ATAC dataset [22]. The demonstration used the Cell Ranger ATAC 2.0.0 fragment file for `atac_pbmc_5k_nextgem` and broad cell annotations from the PBMC5k single-cell embedding. After filtering, 4,437 annotated cells were grouped into B cells (422 cells), monocytes (1,702 cells), and T/NK cells (2,313 cells). All three groups passed the pseudobulk filters and contained 2.1, 10.8, and 13.3 million counted fragments, respectively.

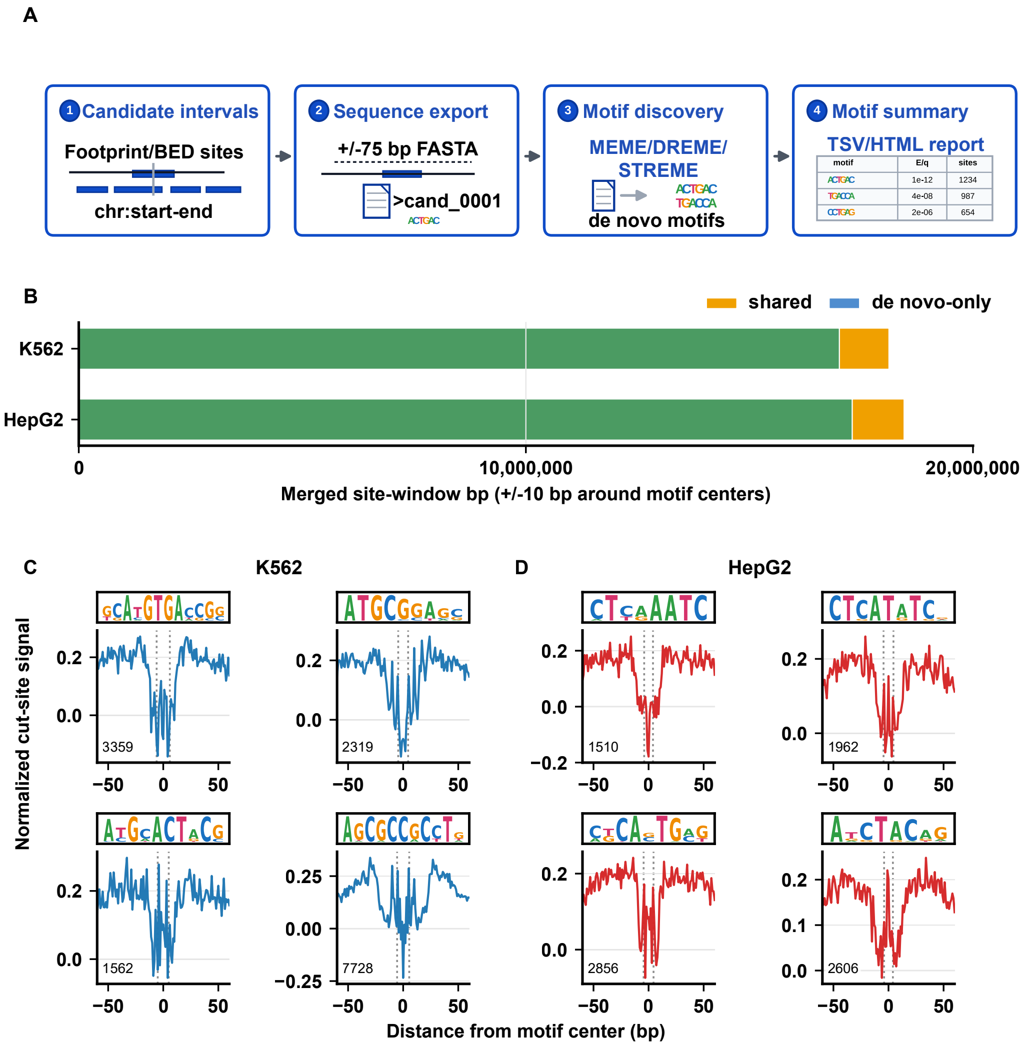

Figure 2. De novo motif discovery and validation summary. (**A**) Candidate footprint or BED intervals are converted into centered FASTA sequences, analyzed with MEME/DREME/STREME, and summarized as motif tables and reports. (**B**) Merged site-window coverage compares shared and de novo-only motif-supported intervals in K562 and HepG2. (**C**,**D**) Representative K562 and HepG2 de novo motifs show footprint-like normalized cut-site aggregate profiles centered on motif instances.

The full `pseudobulk-footprints` route uses the grouped PBMC5k fragments to emit pseudo-paired BAMs, run `atac-correct` with zero read shifting, score corrected footprint tracks, and optionally run motif-aware `diff-footprints` and aggregate plotting. Raw CPM cut-site tracks provide QC and chromatin-context signal alongside corrected footprint scores. Compared with preparing pseudobulk inputs outside the footprinting analysis and then running TOBIAS, `fp-tools` records barcode grouping, pseudobulk QC, correction, footprint scoring, motif-aware comparison, and aggregate plotting in one analysis route. This pseudobulk analysis is complementary to established single-cell chromatin-accessibility ecosystems such as ArchR and Signac and to motif-activity baselines such as chromVAR [23–25].

The same PBMC5k pseudobulk footprint table was used to prioritize marker TFs for cell-level visualization. We summarized the three broad-cell pairwise comparisons as volcano plots and labeled predeclared marker TFs from the two cell types in each comparison only when the differential-footprint direction matched the expected lineage

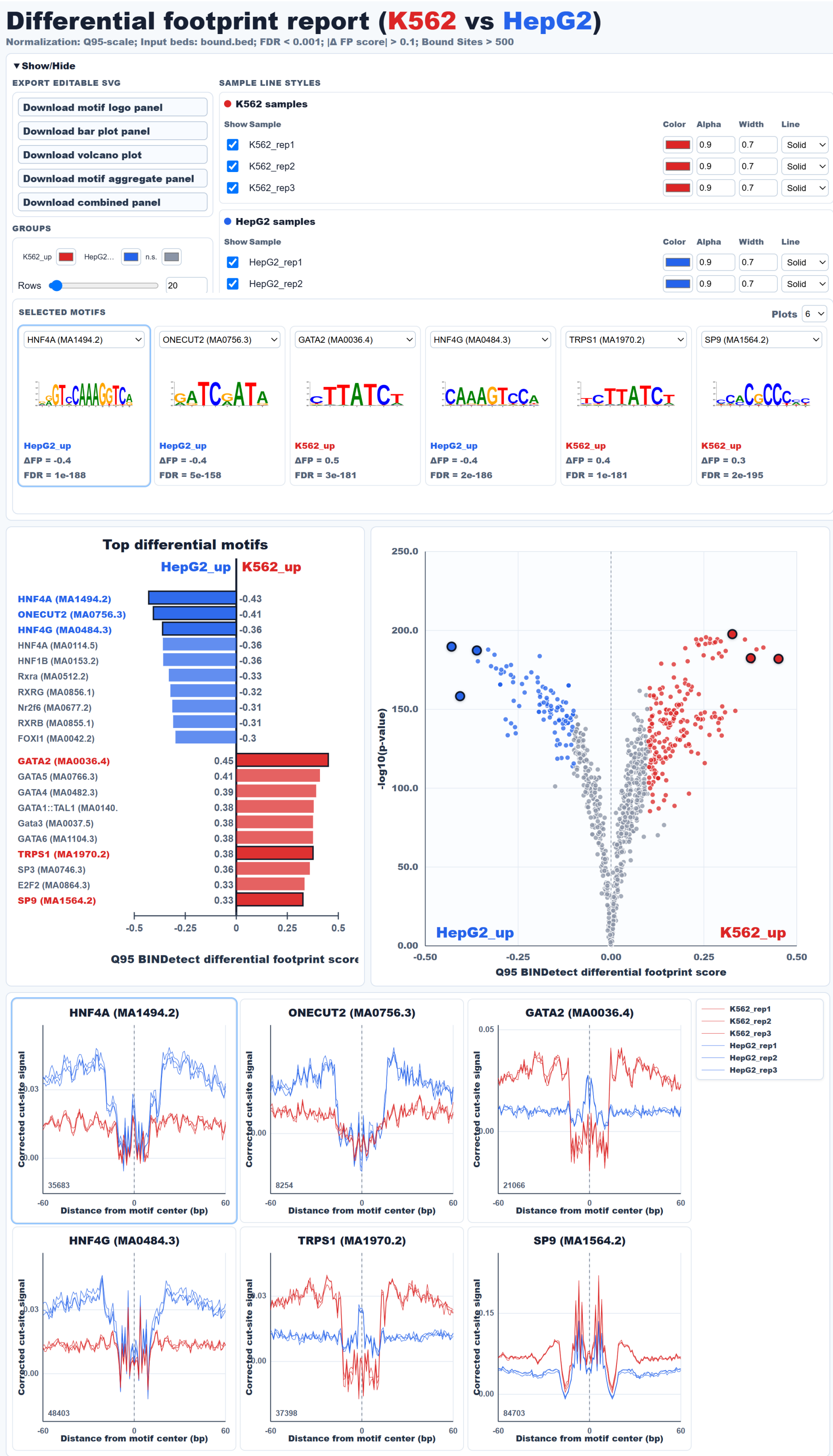

Figure 3. Interactive differential footprint report for K562 versus HepG2. The browser report view summarizes selected differential motifs, motif logos, top differential footprint-score bar plots, a volcano plot, and replicate-level motif-centered aggregate profiles. Blue indicates HepG2-biased footprint scores and red indicates K562-biased footprint scores.

direction. We then projected KNN-smoothed footprint-signature scores onto the PBMC5k UMAP. For each cell, the displayed value summarizes local motif-centered cut-site profiles in neighboring cells. The final figure combines a marker-by-cell heatmap with UMAP panels for broad cell-type labels and representative B-cell-, monocyte-, and T/NK-associated motif signatures (Figure 4).

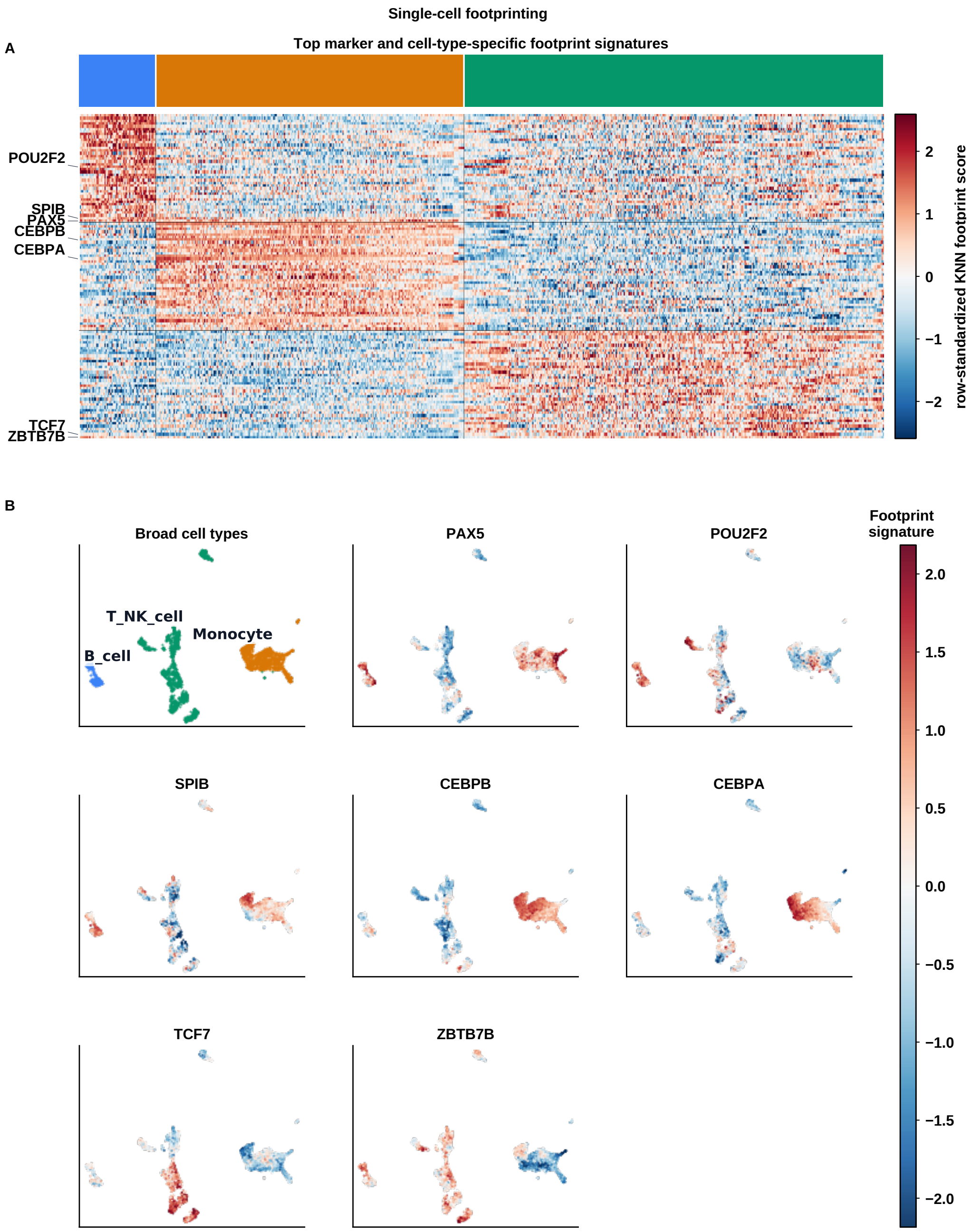

Figure 4. Single-cell footprinting summary in public PBMC5k single-cell ATAC data. (**A**) KNN-smoothed per-cell footprint-signature heatmap for selected marker TFs and top cell-type-specific motif signatures across B cells, monocytes, and T/NK cells. (**B**) UMAP projections show broad cell-type annotations and representative marker footprint-signature z-scores. Positive values indicate the marker-oriented footprint-signature direction after per-TF standardization; ATAC-corrected footprint tracks are computed from grouped pseudobulk profiles.

3.7. Evaluation and Figure Reproducibility

The manuscript analyses are linked to public inputs through manifests, schema checks, recorded command settings, and figure-generation scripts. Together, these files make the bulk, de novo, pseudobulk, replicate-aware, and report examples traceable from input data to final figures, while preserving the sample, motif, region, and signal-track choices used for each analysis.

For each analysis, the workflow writes resolved manifests, software versions, parameters, and plotting tables beside the generated outputs. We also recorded runtime and peak memory for a single-machine comparison of TOBIAS and `fp-tools` using the same input bundle and command outputs (Figure 5). The browser GUI follows the same command structure and provides example loading, YAML review, run history, and report inspection for users who prefer a graphical workflow (Figure 6).

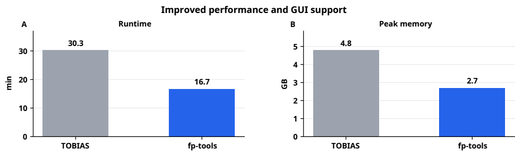

Figure 5. Single-machine runtime and memory check. (**A**) Runtime and (**B**) peak memory are summarized for one local analysis run comparing TOBIAS and `fp-tools` on the same analysis bundle. These measurements are included as an engineering check, not as an exhaustive cross-tool benchmark.

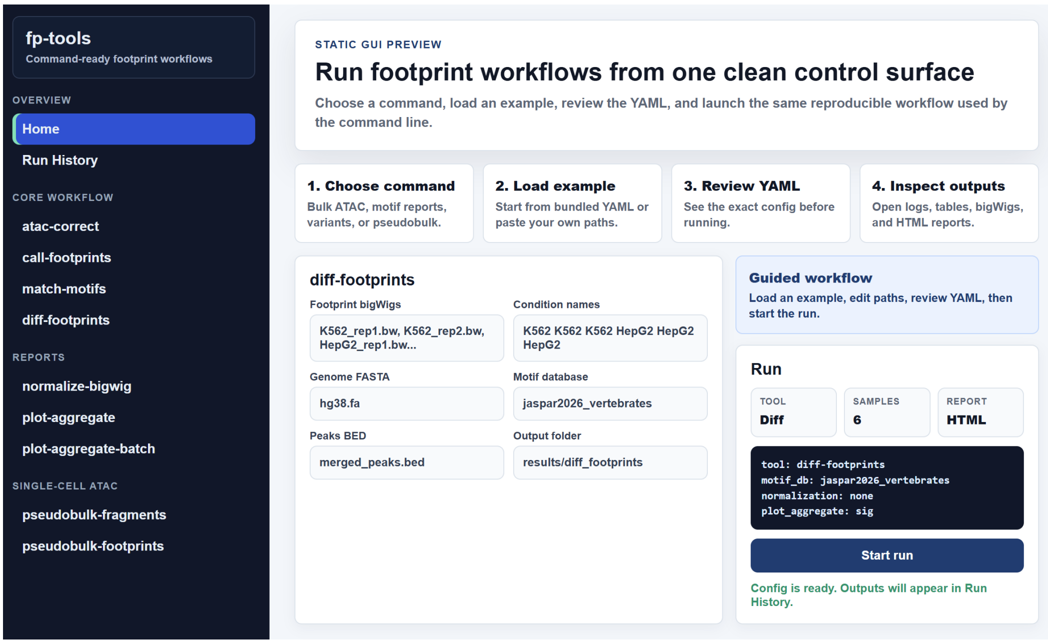

Figure 6. Browser GUI for command-compatible footprinting analyses. The GUI provides sidebar access to core commands, guided configuration panels, example loading, YAML preview, run-history review, and report/output inspection while preserving CLI-compatible configuration files.

4. Discussion

`fp-tools` is a practical extension of motif-centered ATAC-seq footprinting and regulatory motif analysis. Its strengths are consistent interfaces across bias correction, footprint calling, differential footprint analysis, and aggregate visualization; automated output checks; explicit replicate-aware summaries; and integrated de novo motif preparation and pseudobulk aggregation using the same region, motif, and signal conventions (Table 1).

These features address a common practical problem: footprint scoring, motif interpretation, and report generation are often handled by separate scripts or tools.

The closest comparison is TOBIAS, which established a widely used ATAC footprinting workflow for bias correction, footprint scoring, motif-aware differential analysis, and aggregate visualization. `fp-tools` keeps the same interpretable bias-corrected center-versus-flank signal model and improves the surrounding analysis layer: faster core operations, consistent modern command names, reusable YAML execution, report and SVG export, explicit replicate-reporting modes, de novo motif-preparation handoff, pseudobulk single-cell aggregation, GUI access, and tested example outputs.

Table 1. Qualitative scope comparison across widely used ATAC-seq regulatory-analysis tools. ✓, documented native support; ○, partial, indirect, or adjacent support; NA, outside the tool's primary scope. The table summarizes analysis scope from public documentation and primary publications; it is not a performance benchmark and should be read with the different goals of each tool in mind. TOBIAS is the closest core footprinting comparator; the `fp-tools` columns emphasize added analysis and usability components around that motif-centered model.

Feature	fp-tools	TOBIAS	HINT-ATAC	PRINT/ scPrinter	Chrom BPNet	max ATAC
Bulk ATAC footprinting	✓	✓	✓	✓	○	NA
Tn5 bias correction	✓	✓	✓	✓	✓	NA
Footprint calling	✓	✓	✓	✓	NA	NA
Motif-aware differential analysis	✓	✓	○	○	NA	NA
De novo motif preparation	✓	NA	NA	✓	○	NA
scATAC/pseudobulk support	✓	○	○	✓	○	○
Replicate-aware reporting	✓	○	○	○	NA	NA
Visualization/reporting	✓	✓	○	✓	✓	○
YAML/batch execution	✓	NA	NA	NA	NA	NA

Other regulatory-analysis tools address adjacent but distinct questions. Deep sequence-to-profile models such as ChromBPNet learn sequence determinants of chromatin profiles and are well suited for occupancy and variant-effect analyses [10], while accessibility-based frameworks such as diffTF estimate differential TF activity from changes in open chromatin [26]. These approaches are complementary to motif-centered footprinting. The role of `fp-tools` is to make interpretable ATAC-seq footprint analysis easier to run end to end: corrected cut-site signals, motif matching, de novo motif preparation, pseudobulk single-cell analysis, replicate-aware reporting, GUI execution, and publication-ready reports are kept within one reproducible workflow.

5. Conclusions

We presented `fp-tools`, a Python platform that unifies ATAC-seq footprinting with motif-discovery preparation, pseudobulk aggregation, and replicate-aware reporting. By pairing tested commands with public-data manifests, source tables, and reproducible figure generation, the package supports standardized and auditable regulatory-genomics analysis. Future work will add larger whole-genome transfer studies, direct external-tool comparisons, multiscale and nucleosome-aware validation, and stronger hierarchical differential-binding models.

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Institutional Review Board Statement: Not applicable. This study used only publicly available, de-identified datasets and did not involve new experiments on humans or animals.

Informed Consent Statement: Not applicable.

Data Availability Statement: `fp-tools` is openly available in the public GitHub repository `oncologylab/fp-tools`. The repository contains the manifests, schemas, analysis scripts, figure generators, source tables, and small result tables used in this manuscript. Public input data are obtained from the ENCODE Project [18], the 10x Genomics PBMC5k single-cell ATAC example [22], and motif databases such as JASPAR [14] using the included download and preparation scripts. The computational environment is documented in `environment.yml` and the repository Dockerfile.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix

Table S1. Primary analysis parameters used for the public ATAC and pseudobulk workflows. The complete source table is provided in the Supplementary Materials as "Primary analysis parameters".

Step	Parameters
Public input data	Bulk ATAC analyses use public ENCODE K562/HepG2 replicate BAMs and tracks. The pseudobulk single-cell demonstration uses the public 10x Genomics 5k PBMC scATAC fragment dataset. Motif analyses use JASPAR 2026 CORE vertebrate non-redundant motifs.
ENCODE BAM processing	Released GRCh38 alignment BAMs for K562 ENCSR868FGK and HepG2 ENCSR291GJU were filtered to standard chromosomes, indexed, and summarized before footprint analysis.
Blacklist and peak set	Mitochondrial and nonstandard contigs excluded; hg38 blacklist v2 regions subtracted; released ENCODE peak intervals sorted and merged across replicates into one analysis peak set.
Bias correction and footprint scores	`atac-correct` run on filtered BAMs with merged peaks and blacklist regions using default Tn5-bias settings; corrected cut-site tracks q95-scaled over shared background regions for multi-sample analyses; `call-footprints -score footprint` run with default footprint and flank windows.
Differential footprint analysis	The 3-vs-3 ENCODE K562/HepG2 replicate footprint tracks were analyzed with repeated condition labels, JASPAR 2026 motifs, and replicate reporting enabled. One-sample contrasts use motif-site versus background-distribution comparison; replicate-supported analyses use empirical-Bayes variance moderation on per-sample aggregated motif scores.
De novo motif validation	ENCODE K562/HepG2 candidate footprints exported as condition-level FASTA files with $\pm$75 bp flanks; reciprocal STREME 5.5.9 runs used `-dna -nmotifs 250`; Tomtom 5.5.9 matched discovered motifs to JASPAR 2026; de novo motifs were evaluated alone and as additions to known motif sets.
Pseudobulk PBMC analysis	Public 10x PBMC5k scATAC fragments grouped into broad B-cell, monocyte, and T–NK labels, retaining 4,437 annotated cells and requiring at least 300 cells and 50,000 fragments per group. The full `pseudobulk-footprints` route writes pseudo-BAMs, runs ATAC correction with zero read shifting for fragment-derived pseudobulk BAMs, and emits footprint-score tracks, motif-aware differential results, pairwise volcano plots, and KNN-smoothed footprint-signature UMAPs.

Table S2. Software and resource versions used for manuscript analyses. The complete source table is provided in the Supplementary Materials as "Software and resource versions".

Tool or resource	Version	Role
`bedtools`	2.31.1	Region sorting, merging, and blacklist subtraction
Biopython	1.87	Sequence and motif-related utilities
Cell Ranger ATAC	2.0.0	Source processing for the public 10x PBMC5k scATAC dataset
`fp-tools`	0.1.7	ATAC correction, footprint calling, motif matching, differential footprint analysis, pseudobulk utilities, and plotting helpers
hg38 blacklist	v2	Artifact-region filtering for processed bulk ATAC alignments
hg38 reference genome	GRCh38/hg38	Sequence extraction and ATAC bias correction
htslib	1.23.1	SAM/BAM/VCF I/O backend used by `samtools`
JASPAR CORE vertebrates	2026 non-redundant	Known motif database for differential footprint analysis
matplotlib	3.10.9	Manuscript figure rendering
MEME Suite, STREME, and Tomtom	5.5.9	De novo motif discovery and discovered-motif matching
NumPy	2.4.6	Numerical arrays and benchmark/figure calculations
pandas	2.3.3	Tabular processing for benchmark, manifest, and figure builders
pyBigWig	0.3.25	bigWig signal reading and writing
pysam	0.23.3	BAM and genomic interval I/O
Python	3.12.3	Runtime for analyses, figure builders, and provenance extraction
`samtools`	1.23.1	Released ENCODE BAM filtering, indexing, and QC summaries
SciPy	1.17.1	Statistical and interpolation utilities

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